Project Design Phase | Solution Architecture

Date: 27-10-2026 | **Team ID:** SWUID2026403175 | **Project Name:** Agri-Tech

Solution Architecture:

The solution architecture for Agri-Tech ensures a robust, user-friendly, and scalable smart farming platform that connects farmers with weather forecasts, soil analysis, pest alerts, resource ordering, and expert guidance. Built on the MERN stack with a three-tier design, the architecture focuses on responsive and multilingual design, secure role-based access, real-time data delivery, and consistent data management.

- Farm profile, crop monitoring and soil health management in one place
- Real-time weather forecasting and pest alert notifications
- Secure JWT-based authentication with farmer and admin roles
- Resource purchase and order tracking with admin approval
- Expert consultation and community forums for knowledge sharing
- Scalable MongoDB data layer and modular Express.js services